

For the grids A and B given below follow a certain pattern. Analyse A and determine the correct value of '?' in B.

2	11	7
5	13	6
8	13	3

A

7	17	8
10	?	4
5	12	5

B

$$10 + 4 + 2 = 16$$

$$8 + 3 + 2 = 13$$

$$6 + 5 + 2 = 13$$

$$7 + 2 + 2 = 11$$

$$C_2 = C_1 + C_3 + 2$$

3	9	4
8	2	14
3	12	6

A

4	13	5
?	4	15
2	12	6

B

$$3 + 2 + 3 = 8$$

$$4 + 4 + 2 = 10$$

Following a certain pattern, Analyze A and determine the missing value in B.

6	3	2
2	144	4
12	6	1

A

9	3	2
3	270	6
?	5	1

B

Answer Options

Select any one option:

☐ 10

☐ 12

☒ 15

☐ 20

Handwritten notes:

$$6 \times 2 = 12$$

$$12 \times 1 = 12$$

$$12 \times 12 = 144$$

$$(9 \times 2) \times (3 \times 1) = 27$$

$$18x = 270$$

$$x = 15$$

3	4	6	10
10	10	?	8
3	2	3	2
19	2	22	28

Select any one option:

☐ 4

☐ 5

☐ 6

☐ 8

$$(10 \times 2) + 8 = 28$$

$$(6 \times 3) + x = 22$$

$$x = 4$$

1	2	5
2	1	6
3	?	3

8

$$1 + 2 + 5 = 8$$

$$2 + 1 + 6 = 9$$

$$3 + x + 3 = 10$$

$$\Rightarrow x = 4$$

$$x = 4$$

Answer Options

Select any one option

☐ 3

☐ 4

☐ 5

☐ 6

4	2	2
2	82	4
6	1	1

A

3	2	6
3	?	4
1	4	1

B

Answer Options

Select any one option

☐ 102

☐ 92

☐ 82

☐ 72

$$4^2 + 2^2 + 2^2 + 2^2 + 4^2 + 6^2 + 1^2 + 1^2 = 82$$

Similarly

$$3^2 + 2^2 + 6^2 + 3^2 + 4^2 + 1^2 + 4^2 + 1^2 = 92$$

92

35	3	25
45	17	31
39	21	32

57	30	51
62	?	21
36	10	42

A
B

Answer Options

Select any one option

☐ 11

☐ 10

$3 \times 5 - 2 \times 6 = 3$
 $4 \times 5 - 3 \times 1 = 17$
 $3 \times 9 - 3 \times 2 = 21$

$$5 \times 7 - 5 \times 1 =$$

$$6 \times 2 - 2 \times 1 =$$

$$3 \times 6 - 4 \times 2 =$$

10

The two grids A and B given below follow a certain pattern. Analyse A and determine the correct value of T in B.

B	L	J
F	O	I
O	Z	K

H	R	J
S	X	E
T	Y	?

A
B

B = J = L
F = I = O
O = K = Z

similarly, T = Y = S ... so E

$$c_2 = c_1 + c_3$$

Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M
Position value	1	2	3	4	5	6	7	8	9	10	11	12	13
Alphabet	Z	Y	X	W	V	U	T	S	R	Q	P	O	N
Position value	26	25	24	23	22	21	20	19	18	17	16	15	14

$$Y = 7 + ?$$

$$25 = 20 + ?$$

$? = 5$

E

Three flasks of sizes 6 ml, 18 ml, and 24 ml contain a mixture of tonic & water in the ratio 2:1, 4:5, and 7:11, respectively,

The Contents of all the three flasks are poured into a beaker.

Find the ratio of tonic & water in the resultant mixture?

Select any one option

- 17:13
- 13:11
- 11:13
- 13:17

Handwritten solution:

Flask A: 6 ml, Ratio 2:1
 Flask B: 18 ml, Ratio 4:5
 Flask C: 24 ml, Ratio 7:11

Tonic: Water

For A: $\frac{2}{1} \times 6 = 12$ (Tonic), $1 \times 6 = 6$ (Water)
 For B: $\frac{4}{5} \times 18 = 14.4$ (Tonic), $5 \times 18 = 90$ (Water)
 For C: $\frac{7}{11} \times 24 = 15.27$ (Tonic), $11 \times 24 = 264$ (Water)

Total Tonic = 12 + 14.4 + 15.27 = 41.67
 Total Water = 6 + 90 + 264 = 360

Ratio Tonic:Water = $\frac{41.67}{360} = \frac{12.5}{108} = \frac{125}{1080} = \frac{25}{216}$

Final ratio: $\frac{T}{W} = \frac{25}{216}$

What should come in place of the question mark '?' in the following number series?

79, 80, 84, 93, 109, ?

Select any one option

- 134
- 119
- 144
- 121

Handwritten solution:

79, 80, 84, 93, 109, ?

Differences: 1, 4, 9, 16, 25

Next difference: 36

Next number: 109 + 36 = 145

If X + Y means X is the husband of Y, X - Y means X is the wife of Y, X x Y means X is the father of Y, X / Y means X is the mother of Y, X % Y means X is the brother of Y, X * Y means X is the sister of Y, X # Y means X is the son of Y and X @ Y means X is the daughter of Y, then by the given statement, what is the relation of A with P?

A / B % C x K * M # P

Select any one option

- Grandmother
- Daughter-in-law
- Mother
- Mother-in-law

In a row of 120 stalls during a food parade, Ron's stall is 50th from the left side of the row in the same row, Jake's stall is 90th from the right side

From the given scenario, determine the number of stalls between Ron's and Jake's stall

Select any one option

14

12

18

20

You are given three positive integers A, B, and C, and the following relation between them

$$5A = 8B = 6C$$

What is the ratio A: B: C in the given context?

Select any one option

24:15:20

34:56:40

18:29:43

56:30:48

Joey has allocated a budget x for buying gold for his mother's birthday. However, the price of gold increases by 35%.

If Joey decides to spend an additional 20% of x for buying gold then by what percentage will the quantity of gold purchased be reduced?

Note: Assume that the gold is weighed in grams.

Select any one option

11.11%

10.34%

12.33%

14.34%

What is the next number in the given series?

2, 12, 30, 56, 90, ?

Select any one option

121

132

150

145

A watch which gains uniformly is 4 minutes slow at 8 A.M. on Sunday and is 4 minutes 48 sec. fast at 6 A.M. on the next day.

When was the watch showing the correct time?

Select any one option

7:00 pm on Sunday

6:30 pm on Sunday

5:00 pm on Sunday

6:00 pm on Sunday

In a certain system, two positive integers are subtracted in a certain way.
Given below are some examples:

1. $62 - 56 = 0$

2. $48 - 32 = -1$

3. $55 - 31 = 4$

Analyze the given expressions and determine the result of the expression $42 - 28$.

Select any one option

1

2

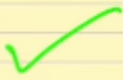
3

4

How many pairs of letters are there in the word "COMPETITION" each of which has as many letters between them as in the English alphabetical series in both forward and backward directions?

Select any one option

- 2
- 3
- 4
- 5



Posi
Posi
A

A person has 3000 bananas and a camel. The person wants to transport the maximum number of bananas to a destination which is 1000 KMs away, using only the camel as a mode of transportation. The camel cannot carry more than 1000 bananas at a time and eats a banana every km it travels.

What is the maximum number of bananas that can be transferred to the destination using only camel (no other mode of transportation is allowed)

533 1

How many consecutive zeros are there at the end of the integer obtained in the product

$$1^2 \times 2^4 \times 3^6 \times 4^8 \times \dots \times 25^{50}?$$

Select any one option

- 50
- 55
- 100
- 200



$$\begin{aligned} 2^4 &= 2^4 \rightarrow 4 \\ 4^8 &= (2^2)^8 = 2^{16} \rightarrow 16 \\ (6)^{12} &= (2 \times 3)^{12} = 2^{12} \rightarrow 12 \\ (8)^{16} &= (2^3)^{16} = 2^{48} \rightarrow 48 \end{aligned}$$

What is the rightmost digit preceding the zeros in the value of 30^{30} ?

Select any one option

- 1
- 3
- 7
- 9



BADMINTON is a nine letter word; its constituent letters are: B, A, D, M, I, N, T, O and N.

Now, if we write their positions in the alphabetical order, the things will look like:

B A D M I N T O N ← Given Words A B Z
2 1 4 13 9 14 20 15 14 ← Alphabet 1 2 26

Now, find the possible matching cases:

A) Cases of matching from forward side:

I) "B * D * * * * *" there is only one letter between B and D in the English alphabet.

B) Cases of matching from reverse side:

I) "B A * * * * *" the letter B comes right after A in the English alphabet.

II) "* * * * * O N" the letter O comes right after N in the English alphabet.

Thus, total three matching pairs could be found.